What is Object Tracking?

Object tracking identifies and follows objects across video frames.

Object Tracking is like giving a machine a “memory” of movement. While **object detection** can answer what’s in this frame, object tracking answers where the same object moves in the next frames and beyond.

How does Object Tracking work

While detection finds objects in a single image, tracking adds a **time dimension.**

1. First, an object detection model (like YOLO) finds objects in the first frame. This gives the initial locations of objects - bounding boxes and labels.

2. In the next frame, the tracking algorithm tries to figure out:

> “Is this object here the same one we saw before?”

This is done using techniques such as:

- motion prediction
- feature similarity
- appearance modeling

Tracking models assign a **unique ID** to each object and then update that ID's position in new frames.

3. Continuous Tracking

The cycle continues detection or prediction, then association, so each object's trajectory through time gets recorded.

This allows analytics like speed, direction, and even behavior patterns.

Types of Object Tracking

- **Single Object Tracking (SOT)** — follows one target (e.g., a racing car).
- **Multiple Object Tracking (MOT)** — follows many targets simultaneously, often with real-world challenges like occlusion (e.g., crowded street scenes).

MOT is especially challenging because:

- objects can disappear and reappear
- objects can look similar
- the scene can be crowded and dynamic

Roboflow and other libraries provide tools to help with multi-object tracking for real applications.

Real World Applications:

- Traffic Monitoring
- Sports Analytics
- Retail Analytics
- Security Systems